



6th Grade Digital Discoveries

Family Overview · Optima Academy Online

~4¼ hrs per week · ~50 min/day,
M–F

About this overview. This document is an example of how the course might be taught. More topics, tools, and projects are approved for this course than appear here, so scholars may see some variation in topics, tools, and projects from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

ABOUT THIS COURSE

A full-year introduction to computing for 6th graders. Scholars build everyday digital communication and design skills, learn how data and the internet really work, write their first programs in Python, and finish the year with a project that ties human-AI collaboration to a real problem.

THE YEAR AT A GLANCE

Unit 1 — Digital Foundations ~7 weeks

Communication, multimedia design, file types, and productivity tools.

Unit 2 — Data & Cyber Safety ~9 weeks

How data moves, phishing & privacy, encryption, and digital wellness; ends with a PSA project.

Unit 3 — Programming ~9 weeks

Computational thinking, flowcharts, and Python basics (variables, conditionals, loops, functions, debugging).

Unit 4 — Modeling, Design & AI ~5 weeks

Simulation, the engineering design process, and the ethics of human-AI collaboration.

Unit 5 — Capstone & Course Final ~2 weeks

Careers in computing and a year-end capstone project.

Capstone & projects. Two flagged projects: a Semester-1 PSA (Unit 2) and a Year-End Capstone (Unit 5).

A TYPICAL WEEK

- One lesson page introducing the week's concept.
- One hands-on activity or assignment (e.g., "Spot the Phish," "Bug Hunter").
- One short knowledge-check quiz.
- Live or VR check-ins at the unit cadence.

TIME COMMITMENT

250 minutes per week (master schedule)

- Lesson reading & video: ~15 min/day
- Activity / assignment work: ~25 min/day
- Quiz or reflection: ~10 min/day

Project and capstone weeks may run longer.

WHAT SCHOLARS NEED

- Optima's VR headset (provided) for VR activities
- Microsoft 365 / OneDrive account
- Microphone for voice-based work
- Reliable internet

AI in this course. AI literacy is part of the course; scholars learn when AI helps and when human judgment must lead.

On-Demand vs. Live. On-Demand (asynchronous): self-paced with due dates for pacing; a teacher is available for support, feedback, and grading.

Live school: live-school scholars take the same course in Optima's VR campus with a teacher and classmates (daily for core courses).

Optima Academy Online · prepared for in-person family meeting

0200022 (M/J Digital Discoveries 1 — FL course code)
Standard middle-school course load — 6th-grade Digital Discoveries was not in the master-schedule export I was given, so the 250 min/week figure follows the school's 6th-grade band.